Prototypical Knowledge Transfer for Multi-Scenario Recommendation with Optimal Transport

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ABSTRACT

Modern APPs often need to provide personalized recommendations across various scenarios, such as the homepage, local page, and live stream on TikTok. User behaviors in these scenarios differ, resulting in diverse data distributions. To effectively handle varying distributions and enhance performance, current Multi-Scenario Recommendation (MSR) methods usually employ parameter sharing for knowledge transfer across scenarios. However, when there are significant differences in scenario distributions, direct parameter sharing fails to achieve effective knowledge transfer and feature alignment. In this paper, we propose a Prototypical Knowledge Transfer (PKT) method with Optimal Transport for Multi-Scenario Recommendation. Specifically, PKT first employs the Multi-gate Mixture of Experts (MMoE) as a shared feature extractor across all scenarios. We then introduce a prototype layer as an intermediary distribution to serve as a bridge for knowledge transfer between different scenario distributions. Furthermore, we leverage Optimal Transport to facilitate efficient knowledge transfer from scenario distributions to prototypes. To extract scenario-specific features, each scenario is equipped with its own expert network and utilizes the LHUC architecture to enhance scenario-specific features. We conduct offline experiments on two datasets and deploy the method on a video platform, followed by online A/B testing. To enhance reproducibility, we have provided the code in an anonymous GitHub repository: https://anonymous.4open.science/r/PTK-78F0.

CCS CONCEPTS

• **Information systems → Recommender systems**.

KEYWORDS

Multi-Scenario Recommendation, Prototype learning, Optimal Transport

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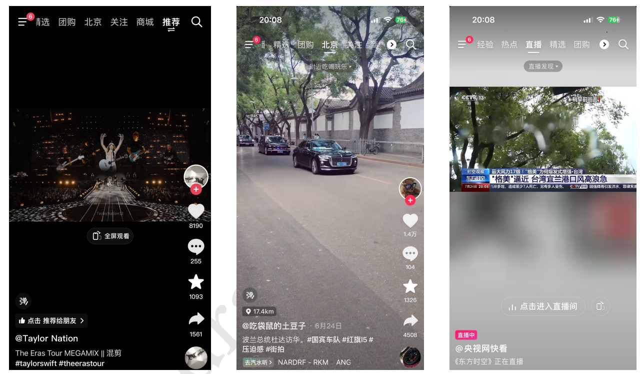


(a) Home page (b) Local page (c) Live page

Figure 1: An example of MSR in Douyin. Different pages, such as the homepage, local page, and live page, correspond to distinct scenarios.

1 INTRODUCTION

Personalized recommendation aims to filter out the items that users are most likely to be interested in from a vast amount of information, thereby addressing the problem of information overload [41, 42]. With the development of the Internet and growing diversity of user needs, the scenarios for providing recommendation services to users have become increasingly diverse [49]. For instance, TikTok includes multiple recommendation scenarios, such as the homepage, local page, and live stream page [48], as shown in Figure 1. While this diversity of scenarios offers users a richer experience, it also presents significant challenges for recommendation systems [45]. In these scenarios, user behavior patterns differ significantly, leading to noticeable differences in the distribution of interaction data across different scenarios [19]. However, establishing a separate model for each scenario neglects common behaviors and interests of users within the APP and increases the maintenance costs for enterprises [2]. Therefore, effectively utilizing multi-source data and improving efficiency have made Multi-Scenario Recommendation (MSR) an important topic in practical applications [11, 43].

Existing Multi-Scenario Recommendation methods typically employ parameter sharing to facilitate knowledge transfer across scenarios [39]. For instance, STAR introduces a scenario-shared model to enable knowledge transfer across various scenarios [1]; SAR-Net and SAML utilize attention layers to facilitate seamless knowledge transfer across different scenarios [30, 46]; AdaptDHM employs shared adaptive modules to differentiate between commonalities and variables of scenarios [22]. However, when there are significant differences in scenario distributions, direct parameter sharing fails to enable effective knowledge transfer and feature alignment [14, 35]. More critically, it can lead to a decline in prediction performance across scenarios. Therefore, designing effective feature alignment [14] and knowledge transfer schemes [10] in Multi-Scenario Recommendation (MSR) is crucial.

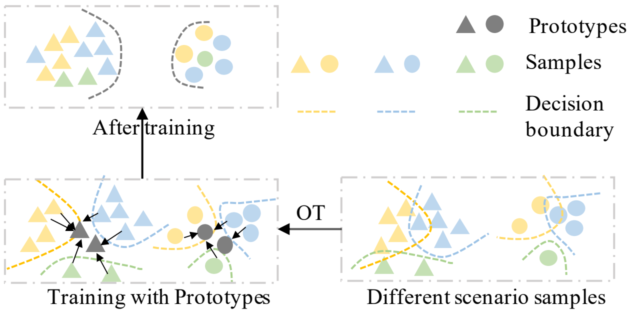


Figure 2: The core idea of our proposed method.

To address the aforementioned issues, in this paper, we propose a Prototypical Knowledge Transfer (PTK) model for Multi-Scenario Recommendation, which integrates Optimal Transport [29]. Figure 2 illustrates the core idea of PTK. Specifically, PTK first employs a Multi-gate Mixture of Experts (MMoE) model [26] as the feature extractor for all scenarios, capturing common features across scenarios and enhance consistency. Next, we introduce a prototype layer as an intermediary distribution space, serving as a bridge between the distributions of multiple scenarios. We design a knowledge transfer scheme based on Optimal Transport, to achieve multi-scenario feature alignment and knowledge transfer by constructing transport targets under a common intermediary distribution. Furthermore, to better capture the unique characteristics of each scenario, we design independent expert networks tailored to each scenario. To further enhance scenario differentiation, we incorporate the Learning Hidden Unit Contributions (LHUC) structure [34] in the expert networks, enabling them to be specifically optimized for the unique demands of their respective scenarios. We conducted comprehensive experiments on two offline datasets, and the results demonstrate that the proposed PTK significantly improves recommendation performance. Additionally, we deployed the PTK on a short video platform for online A/B testing, and the results further validate the method's effectiveness and superiority in practical applications. Our contributions are summarized as follows:

- We propose constructing an intermediary distribution to serve as a bridge for knowledge transfer between different scenario distributions.
- We introduce a Prototypical Knowledge Transfer model (PTK) for Multi-Scenario Recommendation, which utilizes Optimal Transport to efficiently transfer knowledge from scenario distributions to prototypes, thereby improving recommendation performance.
- We conduct offline experiments on two datasets and deploy PTK on a short video platform, demonstrating the effectiveness of our proposed method.

2 RELATED WORK

2.1 Multi-Scenario Recommendation

In recent times, there has been a noticeable shift towards Multi-Scenario Recommendation tasks, driven by the rapid expansion of user bases and web content. Online platforms are increasingly segmenting user groups and content themes into distinct scenarios based on varied attributes. This strategy aligns closely with the principles of multi-task learning, prompting researchers to investigate domain-transfer technologies to address these challenges. One prominent method involves using Mixture-of-Experts (MoE) architectures to manage the diversity across different domains [23, 35, 50]. For instance, the Mario model [35] is designed to capture domain-specific information through feature scaling modules, dynamically adjusting to domain signals using a MoE structure. This model excels in adapting to various domain requirements by leveraging a mixture of specialized expert models. Additionally, HiNet employs a hierarchical structure that constructs multiple layers for information extraction. This approach ensures that while broad domain information is effectively captured, the specific features of each domain are preserved. HiNet's multi-layered framework allows it to balance generalization and specialization, making it highly effective in complex recommendation scenarios. Further innovative methodologies have also been developed. PEPNet [4] processes bottom-level inputs using gating units and integrates EPNet [9] for domain feature selection. This method is complemented by PPNET [16], which combines multi-task information to enhance the overall recommendation system's performance. By integrating multiple layers of domain-specific and multi-task learning capabilities, these models aim to provide more accurate and personalized recommendations.

In recent years, researchers have adopted various innovative approaches to tackle the challenges in domain modeling and adaptation. One such approach involves leveraging dual-structured models like STAR [31], which combines a domain-specific tower with a domain-shared tower to effectively capture and enhance both unique and shared information, thereby improving performance across domains. Attention mechanisms have also been pivotal in advancing domain adaptation. Models such as SAR-Net [30] and SAML [6] utilize attention layers to facilitate seamless knowledge transfer across different domains, resulting in significant performance improvements. Meanwhile, HAMUR [24] focuses on enhancing distribution adaptation through domain adapters, which leads to better prediction outcomes across diverse domains. Prompt-based technologies have found their place in domain adaptation strategies as well. PLATE [40], for instance, employs widely-used NLP prompt techniques to boost adaptation performance. On the other hand, AdaptDHM [22] takes a modular approach, using adaptation modules to differentiate between domain communities and variances, enabling tailored domain strategies for various clusters. Additionally, research efforts such as those by delve into domain knowledge transfer through embedding alignment [27]. CausalInt [38] addresses the refinement of domain recommendations via causal inference, while Adapsarse [44] introduces unique pruning strategies across domains to facilitate further domain adaptation. Recent developments, particularly those embodied in D3 [17], revolutionizes the field by introducing and assessing an autonomous scenario-splitting method, which challenges the traditional manual scenario-splitting approach. Complementarily, the MDRAU [18] leverages knowledge from "seen" domains to aid "unseen" domains. This is achieved through a novel encoder-decoder model named DRIP, which employs "masked domain modeling" strategies to capture preferences at both domain and item levels. The field of

Multi-Scenario Recommendation (MSR) also benefits from new advancements. M-scan [51] integrates two crucial components: a Scenario-Aware Co-Attention mechanism, which extracts user interests from similar scenarios, and a Scenario Bias Eliminator, which applies causal counterfactual inference to mitigate biases. Uni-CTR [13] represents another significant development, utilizing large language models (LLMs) to extract layer-wise semantic representations across different scenarios. Future research in this area could focus on several promising directions. Firstly, refining LLMs for fine-grained scenario alignment is essential, as models like Uni-CTR, while foundational, do not fully extract scenario commonalities, thus limiting scenario expansion. Secondly, expanding the scope of MSR research beyond click-through rate (CTR) tasks to include sequential recommendations and trustworthy recommendations for diverse scenarios remains an underexplored yet vital area. Lastly, the development of joint models that consider multiple tasks, scenarios, behaviors, and interests simultaneously could pave the way for more generalized and robust recommendation systems.

2.2 CTR Prediction

The Click-Through Rate (CTR) is a crucial metric for evaluating user engagement with online content. It provides invaluable insights into user preferences, guiding online platforms in optimizing content strategies to enhance service quality and boost revenue. Accurate CTR prediction is essential for the success of various online services, from advertising to content recommendation.

Traditional CTR prediction models can be broadly divided into four categories, each with distinct characteristics and applications. The earliest attempts at CTR prediction utilized statistical models, with Logistic Regression (LR) [21] being a prominent example. These models are appreciated for their simplicity, interpretability, and ease of implementation, making them a go-to choice for initial CTR prediction efforts. Despite their limitations in handling complex interactions, they provide a solid foundation for understanding basic trends in user behavior. As the need for handling more complex data interactions grew, Factorization Machines (FMs) were introduced [28]. FMs, leveraging matrix factorization techniques, excel in capturing latent relationships in sparse datasets. Their ability to model high-order interactions with linear complexity has made them particularly effective in advertising and recommendation systems, where data sparsity is a significant challenge. With the advent of more powerful computational resources, tree-based models such as Gradient Boosting Decision Trees (GBDT) [12] and XGBoost [5] emerged as potent tools for CTR prediction. These models are adept at handling non-linear relationships and complex feature interactions, which are critical for accurate prediction in dynamic online environments. Their robustness and flexibility have established them as benchmarks in predictive modeling. The latest advancements in CTR prediction involve deep learning models, which offer unparalleled capabilities in modeling intricate feature interactions. DeepFM [15], for example, integrates deep neural networks (DNNs) with FMs to harness the strengths of both approaches. Similarly, Wide&Deep [7] models combine linear models with DNNs to enhance the learning of both memorization and generalization aspects of user behavior. More sophisticated architectures like Deep Cross Networks (DCN) [36], xDeepFM [25], and DCNv2 [37] further push the boundaries by incorporating deep cross networks and low-rank approximations, achieving significant improvements in predictive performance.

3 PRELIMINARY

Optimal Transport (OT) [29] is a mathematical framework designed to transport mass between two distributions with minimal cost. Consider two distributions $\mu = \{x_i, \mu_i\}_{i=1}^{n}$ and $\nu = \{y_j, \nu_j\}_{j=1}^{m}$, where $x_i, y_j \in \mathcal{X}$ represent spatial locations and μ_i, ν_j denote non-negative masses. OT aims to find a transport plan $\pi \in \mathbb{R}_{+}^{n\times m}$ that satisfies $\sum_j \pi_{ij} = \mu_i$ and $\sum_i \pi_{ij} = \nu_j$, ensuring that the row and column marginals match μ and ν. The goal of OT is to minimize the total transport cost, defined by a cost function $C(x, y)$, formulated as:

$$D(\mu,\nu) = \sum_{i,j} \pi_{ij} C(x_i, y_j) = \inf_{\pi\in\Pi(\mu,\nu)} \sum_{i,j} \pi_{ij} C(x_i, y_j), \tag{1}$$

where $D(\mu,\nu)$ represents the optimal transport distance, and $\Pi(\mu,\nu)$ denotes the set of all valid transport plans. This formulation, originally proposed by [32], seeks an optimal coupling $M \in \Pi(\mu_1, \mu_2)$ that minimizes the cost function over the space of joint probability distributions with marginals μ_1 and μ_2. In its discrete form, where μ and ν are empirical measures based on vectors in $\mathbb{R}^d$, the OT distance can be expressed as:

$$OT(\mu_1,\mu_2) = \min_{M\in\Pi(\mu_1,\mu_2)} \langle M, C\rangle, \tag{2}$$

with $\langle\cdot,\cdot\rangle$ denoting the Frobenius dot product and $C \in \mathbb{R}_{\geq 0}^{n\times m}$ being the cost matrix representing pairwise transportation costs between bins i and j. Solving this problem directly is computationally intensive due to the complexity of the constraints. Therefore, entropic regularization [8] is often employed to introduce a penalty term, making the optimization problem more tractable while retaining the fundamental properties of the original OT formulation.

4 METHODOLOGY

In this section, we introduce the proposed method in detail, which is mainly composed of four parts. Section 4.1 provides the formal definition of the Multi-Scenario Recommendation problem. Section 4.2 elaborates on the PTK approach for facilitating information transfer between different scenarios. Section 4.3 introduces the method for extracting scenario-specific features, and Section 4.4 presents the training methodology for PTK.

4.1 Problem Definition

Let $\mathcal{U} = \{u_1, ..., u_{|\mathcal{U}|}\}$, $\mathcal{I} = \{i_1, ..., i_{|\mathcal{I}|}\}$ and $\mathcal{S} = \{1, 2, ..., S\}$ denote the set of users, items and scenarios, respectively, where $|\mathcal{U}|$ is the number of users, $|\mathcal{I}|$ is the number of items, and S is the number of scenarios. The user-item historical interactions are represented by $\mathcal{D} = \{(u, i, s, y) | u \in \mathcal{U}, i \in \mathcal{I}, s \in \mathcal{S}\}$, where $y \in \{0, 1\}$ denotes the binary label. The primary goal of Multi-Scenario Recommendation is to develop a robust scoring function $F(\cdot|\Theta)$, trained on the dataset $\mathcal{D}$, that can accurately predict a user's preference for a given item across various scenarios. Here, Θ represents the set of parameters that define the scoring function F, which determines the likelihood of user u being interested in item i within the specific context of scenario s.

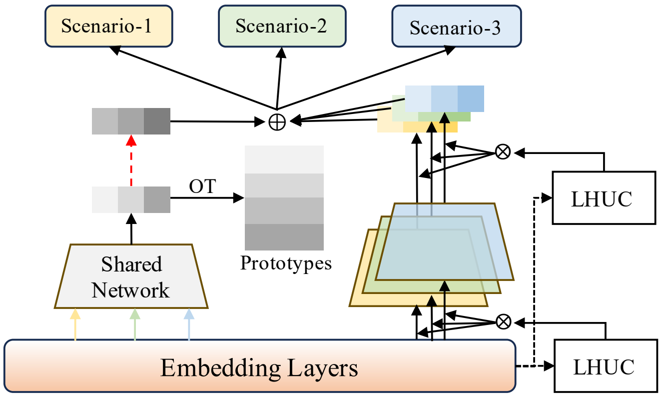


Figure 3: The architecture of PTK.

4.2 Prototypical Knowledge Transfer

Effectively transferring knowledge across multiple scenarios while accounting for differences in data distribution is a critical issue. To address this problem, we propose a Prototypical Knowledge Transfer (PTK) scheme based on Optimal Transport, which facilitates cross-scenario knowledge transfer by aligning the sample distribution with an intermediary distribution. PTK involves three key steps: (1) Designing a shared network to perform basic feature extraction on all scenarios data. (2) Mapping all sample representations to a common prototype space to obtain the projection representations. (3) Utilizing Optimal Transport (OT) to find the minimum distance between the distribution of the projection representations and the original distribution of the sample representations.

4.2.1 ***Shared Feature Extraction Network.*** We first utilize a shared feature extraction network to capture common representations across multiple scenarios. This is based on the Multi-gate Mixture of Experts (MMoE) architecture [26], which allows for flexible parameter sharing and consistency of different scenarios. The common representation of i-th sample under scenario s is:

$$\mathbf{h}_i^s = \sum_{m=1}^{M} g_m^s(x_i) * Q_m(x_i), \tag{3}$$

where $Q_m(x)$ denotes the m-th expert network, which is composed of multi-layer perceptron (MLP) with ReLU activation function. Here, M represents the total number of expert networks. And $g^s(x)$ denotes the gating network for scenario s, which is a simply linear transformation of input x_i with a softmax layer:

$$g^s(x_i) = softmax(W_g x_i), \tag{4}$$

where W_g is the weight matrix for the gating network. The use of the MMoE [26] architecture ensures that each scenario benefits from shared knowledge.

4.2.2 ***Representation Projection with Prototypes.*** Although shared networks can project features from different scenarios into the same semantic space, distribution drift between different scenarios often occurs, and simple feature alignment may lead to negative transfer. To address this issue, we propose to use prototypes as an intermediary distribution to serve as a bridge for knowledge transfer and feature alignment across different scenarios. Let $\mathbf{P} = \{\mathbf{p}_k\}_{k=1}^{K} \in R^{K*d}$ be a set of prototype vectors, where K is the number of prototypes and $\mathbf{p}_k$ is the representation of k-th prototype with d as the hidden dimension. Our aim is to project the sample representation $\mathbf{h}^s$ of different samples into the same prototype space $\mathbf{P}$. This process involves calculating the weight vector λ using a softmax function over a linear transformation of $\mathbf{h}^s$:

$$\lambda = softmax(W_c \mathbf{h}^s), \tag{5}$$

where W_c is the trainable weight matrix. The final projection representation $\mathbf{h}^p$ is then obtained by a weighted sum of the prototypes:

$$\mathbf{h}^p = \sum_{k=1}^{K} \lambda_k * \mathbf{p}_k \tag{6}$$

4.2.3 ***Multi-scenario Optimal Transport.*** To build the bridge between the sample distribution and the intermediary distribution, we use the Optimal Transport (OT) [29] method to construct the distribution transition. Let $\mu_i = \delta(\mathbf{h}_i^s)$, $\upsilon_i = \sum_{k=1}^{K} \lambda_k * \delta(\mathbf{p}_k)$ represent discrete distributions of sample representation $\mathbf{h}^s$ and projection representation $\mathbf{h}^p$, respectively, where δ is the Dirac function. OT aims to find the matching matrix that minimizes the distance between the two distributions. For discrete distributions, given the cost between individual elements, this can be formalized as follows:

$$\begin{aligned} \mathrm{OT}(\mu_i, \upsilon_i) &= \min_{\mathbf{T}_i \in \Pi(\mu_i, \upsilon_i)} \langle \mathbf{T}_i, \mathbf{C}_i \rangle \\ &= \min_{\mathbf{T}_i \in \Pi(\mu_i, \upsilon_i)} \sum_{k=1}^{K} \mathbf{T}_{ik} \mathbf{C}_{ik} + \epsilon \sum_{k=1}^{K} \mathbf{T}_{ik} \log \mathbf{T}_{ik}, \end{aligned} \tag{7}$$

where $\langle \cdot, \cdot \rangle$ is the Frobenius dot-product and $\mathbf{C}$ is the cost matrix with each element calculated as $\mathbf{C}_{ik} = 1 - \cos(\mathbf{h}^s, \mathbf{p}_k)$, representing the distance between the i-th sample representation and the k-th prototype vector. $\mathbf{T}$ is the transport probability matrix, which is learned by minimizing $\langle \mathbf{T}, \mathbf{C} \rangle$. Directly optimizing the first term of Eq(7) often comes at the cost of heavy computational demands, and OT with entropic regularization [8], i.e. the second term of Eq(7), is introduced to allow the optimization at small computational cost in sufficient smoothness. And ϵ is the strength of entropic regularization, the more ϵ increases, the distribution $\mathbf{T}$ tends to become more uniform. The resulting system can be efficiently solved using the classical Sinkhorn divergence algorithm [8].

Since the OT matching process is fully differentiable, we can train our entire network end-to-end. In other words, the average OT distance between μ_i and υ_i over total training sets could be treated as loss function to be optimized:

$$\begin{aligned} \min \mathcal{L}_{ot} &= \min \frac{1}{N} \sum_{i=1}^{N} \mathrm{OT}(\mu_i, \upsilon_i) \\ &= \min \frac{1}{N} \sum_{i=1}^{N} \min_{\mathbf{T}_i \in \Pi(\mu_i, \upsilon_i)} \langle \mathbf{T}_i, \mathbf{C}_i \rangle. \end{aligned} \tag{8}$$

Optimizing Eq.(8), we drive the sample distribution μ towards the intermediary distribution υ, encouraging each prototype $\mathbf{p}$ to capture the common information across all scenarios, which facilitates knowledge transfer and feature alignment in MSR.

4.3 Specific Scenario Expert Module

While PTK captures the common patterns across all scenarios with OT, but each scenario may have its own perspective on the input information. Based on this idea, our scenario expert module establishes S expert networks $\mathcal{S} = \{f^s(\cdot)\}_{s=1}^{S}$, where $f^s(\cdot)$ represents the expert for each scenario, to explicitly handle scenario-specific information. The network architecture is similar to MMoE [26], but each expert network has independently learnable parameters tailored to its specific scenario.

Additionally, we introduce a gating mechanism known as Learning Hidden Unit Contributions (LHUC) [34], which is widely used in the field of speech recognition. LHUC allows for the injection of personalized prior information into the network. It focuses on learning the contributions of hidden units specific to individual speakers, thereby adjusting the model's hidden layers using personalized contributions to improve the accuracy of speech recognition for different speakers. We feed the input x into LHUC, a two-layer neural network. The first layer performs feature crossing:

$$x^{fc} = ReLU(Wx + b). \tag{9}$$

The second layer generates custom gating scores:

$$\kappa = 2 * \sigma(W^{fc}x^{fc} + b^{fc}), \tag{10}$$

where W, W^{fc}, b and b^{fc} are learnable weights. ReLU is chosen as the non-linear activation function. $\sigma(\cdot)$ is the sigmoid function. As shown in Eq(9) and Eq(10), LHUC uses the prior information x to generate personalized gating κ, adaptively controlling the importance of the prior information. Additionally, a constant is multiplied to further compress and amplify the effective signal.

We integrate LHUC into the input and output layers of each expert $f^s(\cdot)$ to enhance scenario-specific features. For a sample x_i belonging to scenario s, its specific representation is calculated as follows:

$$\mathbf{h}^e = \kappa_{out} \otimes f^s(x_i \otimes \kappa_{in}), \tag{11}$$

where κ_{in} and κ_{out} are the gating scores of x_i and $f^s(x_i)$, respectively. And $\otimes$ denotes the element-wise product.

4.4 Training

Finally, we concatenate the projection representation $\mathbf{h}^s$ and the specific representation $\mathbf{h}^e$ of sample x, and input them into the classifier to obtain the final prediction value:

$$\hat{y} = \sigma(\mathbf{h}^p \oplus \mathbf{h}^e) \tag{12}$$

We use binary cross-entropy loss to measure the prediction error:

$$\mathcal{L}_{bce} = \frac{1}{N}\sum_{i=1}^{N} y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i), \tag{13}$$

where N is the size of training set.

The total loss function incorporates both the binary cross-entropy loss and the optimal transport loss:

$$\mathcal{L}_{total} = \mathcal{L}_{bce} + \alpha * \mathcal{L}_{ot}, \tag{14}$$

where α is a hyper-parameter. Algorithm 1 shows the training process of PTK.

Algorithm 1 Training process

Require: Training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, deep recommendation model $F(\cdot;\Theta)$

while Θ has not converged **do**
  Sample mini-batch of size n_b from $\mathcal{D}$
  for $i \leftarrow 1$ to n_b **do**
    Compute sample common representation $\mathbf{h}_i^s$ for x_i
    Obtain sample representation distribution $\mu_i = \delta(\mathbf{h}_i^s)$
    Compute i-th sample projection representation weight $\lambda_i = softmax(W_c\mathbf{h}_i^s)$
    Compute sample projection representation $\mathbf{h}_i^p = \sum_{k=1}^{K} \lambda_k * \mathbf{p}_k$
    Obtain sample projection representation distribution: $\upsilon_i = \sum_{k=1}^{K} \lambda_k * \delta(\mathbf{p}_k)$
    Calculate the OT distance: $\mathrm{OT}(\mu_i, \upsilon_i) = \min_{\pi_i \in \Pi(\mu_i,\upsilon_i)} \langle \mathrm{T}_i, \mathrm{C}_i \rangle$
    Compute sample specific representation $\mathbf{h}_i^e$ for x_i
    Compute prediction $\hat{y}_i = \sigma(\mathbf{h}_i^p \oplus \mathbf{h}_i^e)$
  end for
  Average the OT distance within the mini-batch to compute $\mathcal{L}_{\mathrm{ot}} = \frac{1}{n_b}\sum_{i=1}^{n_b} \min_{\pi_i \in \Pi(\mu_i,\upsilon_i)} \langle \mathrm{T}_i, \mathrm{C}_i \rangle$
  Compute $\mathcal{L}_{\mathrm{bce}} = \frac{1}{n_b}\sum_{i=1}^{n_b} \mathcal{L}(\hat{y}_i, y_i)$
  Update Θ by minimizing $\mathcal{L}_{total} = \mathcal{L}_{\mathrm{bce}} + \alpha * \mathcal{L}_{\mathrm{ot}}$ through gradient descent
end while
return $F(\cdot;\Theta)$

Table 1: Statistics information of datasets.

Dataset	Movielens			Douban		
Domain	D1	D2	D3	D1	D2	D3
#user	1325	2096	2619	2212	1820	2712
#item	3429	3508	3595	95872	79878	34893
#interaction	210747	395556	393906	227251	179847	1278401

5 EXPERIMENT

In this section, we conduct extensive experiments to evaluate PTK, with the purpose of answering the following questions:

- **Q1**: How does the proposed method perform compared with state-of-the-art MSR models?
- **Q2**: What is the effect of different components in the proposed method?
- **Q3**: How does PTK perform in real-world applications?

5.1 Experimental Settings

5.1.1 ***Datasets.*** We evaluate the effectiveness of PTK on two public benchmark recommendation datasets, i.e., MovieLens-1M and Douban. The statistics are shown in Table 1. The split ratio of training, validation, and test is 8:1:1.

- Movielens: The MovieLens dataset is an extensive repository of movie ratings and associated information, frequently utilized in research and the development of recommender systems. This dataset includes user ratings, demographic

details, movie metadata, and user preferences. Specifically, it comprises 1 million anonymous ratings for approximately 4,000 movies, contributed by 6,000 MovieLens users. As recommender systems have evolved, the MovieLens dataset has become an invaluable resource, providing insights into user preferences and supporting the creation of innovative recommendation algorithms, ultimately benefiting movie enthusiasts around the globe. In our MSR task, we aim to perform a multi-scenario evaluation by categorizing interaction samples into three distinct scenarios based on the "age" attribute: "1-24", "25-34", and "35+".
- Douban [47]: The Douban dataset, sourced from the Douban platform, is organized into three distinct subsets: book, music, and movie. These subsets share a common user base, and each subset is treated as a separate scenario in our analysis. User features such as "living place" and "user ID" are preserved. For item identification, all items across the three domains are systematically renumbered and assigned new IDs. According to previous research, ratings above 3 are categorized as positive, while ratings of 3 or below are categorized as negative.

5.1.2 ***Baselines.*** We use the following state-of-the-art MSR/MTR model as baselines:

- SharedBottom [3]: The "Shared Bottom" model is an architectural approach in multi-task recommendation systems that extracts common representations. It has been adapted for Multi-Scenario Recommendations, serving as a strong baseline by treating each scenario as a unique recommendation task.
- MMOE [26]: The Multi-gate Mixture-of-Experts (MMOE) model is a multi-task learning framework that integrates expert networks to handle diverse tasks. It uses gating networks to dynamically distribute task-specific knowledge, improving adaptability in complex scenarios.
- SAR-Net [30]: The Scenario-Aware Ranking Network (SAR-Net), developed by Alibaba, is a recommendation system for travel marketing that integrates diverse data sources and mitigates bias in promotional data. It has been validated and deployed across Alibaba's platforms.
- STAR [31]: The Star Topology Adaptive Recommender (STAR) model optimizes click-through rates across diverse marketing scenarios by combining a shared network with domain-specific adaptations. Its deployment in Alibaba's advertising system has significantly boosted performance metrics.
- AdaptDHM [22]: The Adaptive Distribution Hierarchical Model is an innovative multi-distribution approach that focuses on click-through rate (CTR) prediction across multiple domains. Characterized by an end-to-end hierarchical architecture, it encompasses both a clustering and a classification mechanism. The pivotal element of this model is the distribution adaptation module, which incorporates a routing strategy to dynamically assign each sample to a distribution cluster.
- EPNet [4]: PEPNet, combining PPNet and EPNet, excels in multi-task recommendation by personalizing neural network parameters and user-specific embeddings. It achieves enhanced performance through scenario-specific customization.

5.1.3 ***Evaluation Metrics.*** To evaluate the performance of our proposed model, we utilize two widely-used metrics in Top-n recommendations:

- RECALL@n is a measure of how many of the relevant items are successfully retrieved in the Top-n list.
- NDCG@n (Normalized Discounted Cumulative Gain) is a measure that considers both the relevance and the ranking of the retrieved items. It assigns higher scores to relevant items that appear higher in the recommendation list.

We report the results on two datasets when $n = 3$ and $n = 5$.

5.1.4 ***Hyper-parameter and training details.*** The input embedding dimension is set to 32 for all methods. In shared feature extraction network, the $Q(\cdot)$ is a three-layer MLP witg activation function of ReLU, and M is set to 3. The number of prototype K is searched in $\{8, 16, 32, 64\}$. The hyper-parameter of training, α and ϵ, which control the dregree of OT loss and entropic regularization, respectively, are both searched in the range of $\{0.1, 0.2, ..., 1.0\}$. We optimize all models with Adam [20] optimizer with batch sizes of $\{128, 256, 512\}$. We use grid search to find the optimal hyperparameters. The learning rate is searched in $\{1e-3, 1e-4, 1e-5\}$. To avoid overfitting, we set the dropout [33] to 0.5 and the patience of earlystop to 5 epochs. We conduct all experiments with tensorflow 2.7.0 and Python 3.7.

5.2 Overall Performance

We compare PKT with baseline models on two datasets, as shown in Tab 2 and 3. The experimental results indicate that PKT outperforms the baselines across all metrics and scenarios. From these results, we can safely draw the following conclusions: Multi-Scenario Recommendation models (such as SAR-Net, STAR) outperform simple parameter-sharing models (such as SharedBottom and MMoE) in overall performance. When there are significant differences in data distribution between scenarios, traditional parameter-sharing methods can result in poor knowledge transfer and even negative transfer issues. Multi-Scenario Recommendation models achieve cross-scenario knowledge sharing through different strategies. PKT overcomes these challenges by introducing a prototype layer as an intermediary distribution and utilizing Optimal Transport (OT) for cross-scenario knowledge sharing and feature alignment. The OT method ensures efficient knowledge transfer and alignment between different scenarios by minimizing the transport distance between the sample distribution and the intermediary distribution. This approach not only outperforms other baseline models in overall performance but also excels in handling scenarios with significant distribution differences. Furthermore, PKT enhances the ability to capture scenario-specific features by designing scenario-specific expert networks and incorporating the LHUC structure. This ensures that each scenario benefits from the intermediary distribution while maintaining its unique feature representation. In the Movielens dataset, the scenario distribution is relatively uniform, whereas, in the Douban dataset, there are significant differences between scenario distributions. Nevertheless, PKT effectively handles

Table 2: The overall top-n performance of different methods on Movielens. R@n and N@n denote RECALL@n and NDCG@n, respectively. Bold scores indicate the best in each column, underlined scores indicate the best baseline. We report the results of different methods both overall and across various scenarios. For all metrics, the higher the result, the better.

	Overall		Scenario-1		Scenario-2		Scenario-3		Overall		Scenario-1		Scenario-2		Scenario-3	
	R@3	N@3	R@3	N@3	R@3	N@3	R@3	N@3	R@5	N@5	R@5	N@5	R@5	N@5	R@5	N@5
SharedBottom	0.4586	0.3721	0.4615	0.3705	0.4294	0.3605	0.4813	0.3842	0.5984	0.4590	0.6014	0.4594	0.5657	0.4461	0.6238	0.4713
MMOE	0.4642	0.3783	0.4644	0.3772	0.4356	0.3672	0.4871	0.3878	0.6030	0.4632	0.6087	0.4633	0.5719	0.4522	0.6251	0.4719
SAR-Net	0.4623	0.3787	0.4604	0.3764	0.4338	0.3671	0.4861	0.3884	0.6018	0.4637	0.6059	0.4626	0.5702	0.4521	0.6250	0.4735
STAR	0.4613	0.3762	0.4645	0.3776	0.4317	0.3640	0.4833	0.3853	0.5994	0.4607	0.6078	0.4636	0.5653	0.4475	0.6225	0.4697
AdaptDHM	0.4595	0.3737	0.4605	0.3714	0.4298	0.3616	0.4826	0.3845	0.5994	0.4587	0.6021	0.4562	0.5677	0.4472	0.6234	0.4693
EPNet	0.4701	0.3798	0.4728	0.3808	0.4361	0.3660	0.4943	0.3891	0.6086	0.4631	0.6181	0.4679	0.5698	0.4499	0.6315	0.4762
PTK	**0.4777**	**0.3882**	**0.4818**	**0.3887**	**0.4451**	**0.3742**	**0.5001**	**0.3982**	**0.6201**	**0.4747**	**0.6308**	**0.4775**	**0.5835**	**0.4597**	**0.6442**	**0.4853**

Table 3: The overall top-n performance of different methods on Douban.

	Overall		Scenario-1		Scenario-2		Scenario-3		Overall		Scenario-1		Scenario-2		Scenario-3	
	R@3	N@3	R@3	N@3	R@3	N@3	R@3	N@3	R@5	N@5	R@5	N@5	R@5	N@5	R@5	N@5
SharedBottom	0.2077	0.2397	0.5408	0.3890	0.5954	0.4151	0.2438	0.2671	0.2928	0.3120	0.6531	0.4563	0.6963	0.4784	0.3408	0.3454
MMOE	0.2093	0.2407	0.5394	0.3870	0.5946	0.4146	0.2443	0.2674	0.2925	0.3119	0.6505	0.4539	0.6956	0.4775	0.3407	0.3451
SAR-Net	0.2091	0.2413	0.5415	0.3893	0.5993	0.4180	0.2434	0.2673	0.2931	0.3132	0.6526	0.4561	0.6977	0.4805	0.3414	0.3461
STAR	0.2074	0.2390	0.5377	0.3868	0.5965	0.4141	0.2429	0.2654	0.2923	0.3106	0.6532	0.4555	0.6957	0.4768	0.3412	0.3441
AdaptDHM	0.2098	0.2412	0.5399	0.3879	0.5981	0.4152	0.2451	0.2678	0.2926	0.3122	0.6495	0.4531	0.6949	0.4761	0.3401	0.3453
EPNet	0.2091	0.2409	0.5385	0.3877	0.5962	0.4143	0.2450	0.2677	0.2927	0.3121	0.6539	0.4563	0.6966	0.4773	0.3399	0.3446
PTK	**0.2108**	**0.2426**	**0.5696**	**0.4075**	**0.6194**	**0.4309**	**0.2481**	**0.2717**	**0.2949**	**0.3144**	**0.6892**	**0.4785**	**0.7233**	**0.4962**	**0.3453**	**0.3507**

Table 4: Ablation results with overall performance.

Dataset	Movielens		Douban	
Overall	R@3	N@3	R@3	N@3
w\o OT	0.4764	0.3875	0.2084	0.2411
w\o H(T)	0.4773	0.3880	0.2105	0.2425
w\o prototypes	0.4728	0.3848	0.2061	0.2389
w\o LHUC	0.4768	0.3876	0.2093	0.2408
PTK	**0.4777**	**0.3882**	**0.2108**	**0.2426**

recommendation tasks across different scenarios in both cases. PKT shows significant improvements in Recall@3 and NDCG@3 for all scenarios in the Movielens dataset, and its performance is particularly outstanding in Scenario-1 and Scenario-2 of the Douban dataset. This demonstrates PKT's strong adaptability and robustness in addressing differences in scenario distributions.

5.3 Ablation Study

To further investigate the effectiveness of PTK, we conduct ablation studies by removing key components individually. Specifically, we present the results of the following variants:

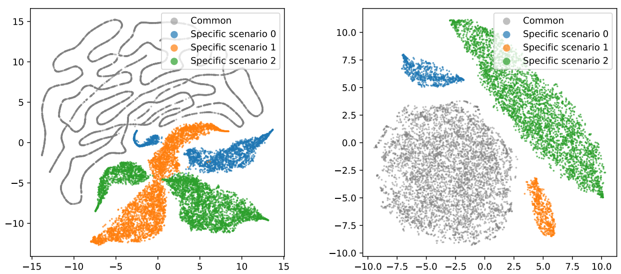
- w\o OT refers to not using OT to align sample common representation $\mathbf{h}^s$ and sample projection representation $\mathbf{h}^\mathbf{p}$.
- w\o $H(\mathbf{T})$ refers to not using entropic regularization.
- w\o prototypes refers to only using Shared Feature Extraction Network to learn common representations $\mathbf{h}^s$.
- w\o LHUC refers to deleting LHUC in Specific Scenario Expert Module.

From Tab 4, we can draw the following conclusions: Removing OT results in decreased performance on two datasets. This indicates that OT plays a crucial role in aligning the common and projection representations of samples, effectively facilitating knowledge sharing and feature alignment. The slight performance drop when entropic regularization is not used suggests that it helps smooth the optimization of the transport matrix. The significant performance decline after removing the prototypes indicates that the prototype layer, serving as an intermediary distribution, successfully achieves knowledge transfer and feature alignment across different scenarios, greatly enhancing the overall model performance. Lastly, the performance decrease after removing LHUC demonstrates that the LHUC is essential for capturing scenario-specific features and enhancing model performance.

5.4 Visualization

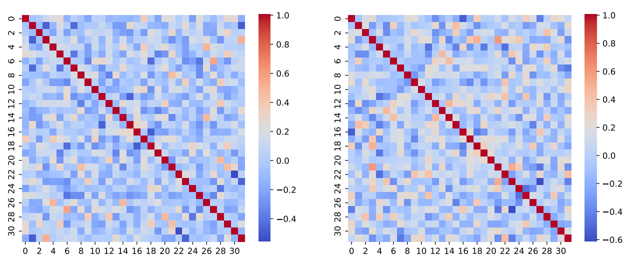
To provide an overview of the effectiveness of PTK in multi-scenario learning, we visualize the sample projection representation $\mathbf{h}^p$ and sample specific representation $\mathbf{h}^e$. Fig4 (a) and (b) show the distribution of samples in the prototype space and the specific scenario spaces on the Movielens and Douban datasets, respectively. It can be seen that PKT effectively leverages the intermediary distribution $\mathbf{P}$ to learn common knowledge across samples from different scenarios, while the scenario specific expert modules retain the unique features of each scenario. Moreover, in Movielens, the data distribution across different scenarios is relatively uniform, whereas in Douban, the data distribution varies significantly between scenarios. Nevertheless, PKT effectively works in both cases, demonstrating its capability to achieve multi-scenario knowledge transfer and feature alignment through the combination of OT and prototypes to improve prediction performance.

Additionally, we provide heatmaps of the prototype cosine similarity, as shown in Figures 5 (a) and (b), illustrating the similarity distribution among the prototypes. The redder the color, the higher the similarity between prototypes; the bluer the color, the lower the similarity. It shows that almost all prototypes are dissimilar to each other, indicating that the prototypes, as an intermediary



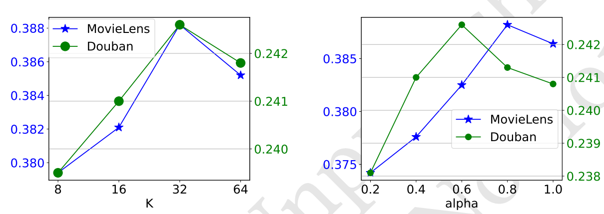
(a) Movielens (b) Douban

Figure 4: The distribution of common and specific representations of samples was visualized by using t-SNE for dimensionality reduction.



(a) Movielens (b) Douban

Figure 5: The heatmap of the prototypes cosine similarity.



(a) K (b) α

Figure 6: Hyper-parameter K and α analysis results.

distribution, effectively absorb and store information from different scenarios.

5.5 Hyper-Parameter Analysis

We analyzed two key hyper-parameters in the PTK model, K and α, to evaluate their impact on model performance. Fig 6 shows the NDCG@3 under different hyper-parameter settings.

In Figure 6(a), we analyze the impact of different numbers of prototypes K on model performance. As shown in the figure, the model performance initially increases with the value of K, but then decreases. On both the Movielens and Douban datasets, the model performance peaks when K is 32. This indicates that an appropriate number of prototypes helps better capture the common and specific information across different scenarios, while too many prototypes may increase the model complexity and negatively affect performance.

Figure 6(b) shows the impact of different weight coefficients α on model performance. We found that as α increases, the model performance gradually improves, reaching its peak. This indicates that an appropriate weight coefficient can better balance the binary cross-entropy loss and the optimal transport loss, thereby enhancing the model's recommendation performance.

Table 5: Online A/B results in three representative scenario in our APPs. We show the relative performance of PTK. The grey square brackets represent the 95% confidence intervals for online metrics.

Scenario	Watch Time	Like	Comment
Discovery page	+0.138% [0.01%, 0.29%]	+0.092% [−0.12%, 0.30%]	0.037% [−0.30%, 0.38%]
Home page	+0.658% [0.06%, 1.26%]	+0.296% [−0.11%, 0.70%]	+0.587% [−0.34%, 1.49%]
Slide page	+0.502% [0.05%, 1.05%]	+0.030% [−0.09%, 0.15%]	+0.322% [0.05%, 0.59%]

5.6 Online A/B Test

To further validate the effectiveness of PTK in practical applications, we conducted online A/B tests in three representative scenarios within the application: the Discovery page, the Home page, and the Slide page. The test results show that the PTK method significantly improved viewing time, likes, and comments in all scenarios. This indicates that PTK can effectively increase user viewing time and interaction frequency, further demonstrating its advantages and robustness in Multi-Scenario Recommendation tasks. These improvements not only confirm the effectiveness of PTK in cross-scenario knowledge transfer and feature alignment but also showcase its broad potential and adaptability in real-world applications.

6 CONCLUSION

In this paper, we introduced a Prototypical Knowledge Transfer (PKT) method with Optimal Transport for Multi-Scenario Recommendation. Our approach starts by employing the Multi-gate Mixture of Experts (MMoE) as a shared feature extractor across all scenarios. We then introduce a prototype layer as an intermediary distribution to facilitate knowledge transfer between different scenario distributions. Optimal Transport is utilized to achieve efficient knowledge transfer from scenario distributions to prototypes. Additionally, each scenario is equipped with its own expert network using the LHUC architecture to enhance scenario-specific features. Our extensive offline experiments on two datasets and subsequent online A/B testing on a video platform demonstrated that the PKT method effectively handles diverse data distributions and significantly enhances recommendation performance across various scenarios.

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